Lavoisier's ghost challenge Data Sheet 1-6

Date____

Your name____________________________________

Partner's name_________________________________

TA____

Make up TA (where applicable)____________________

1

Data Sheet 1. Station 1.

Toledo Electronic Balance

- Mass of exactly 1 ounce of NaCl weighed with 3 decimal balance +/- reading uncertainty

_____**+/-**______

_____**+/-**______

_____**+/-**______

Mean (average) mass (units)

________ **(____)**

SD (units)

________ **(____)**

Number of sig figs in the average mass

____

Calculations:

SD = $\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$

2

Data Sheet 1. Station 1.

Toledo Electronic Balance

Mean mass +/- SD

______+/-_____(___)

→

%RSD

______+/-_____(___)

→

Mean mass +/- %RSD

______+/-____ %

Compare your SD to the balance tolerance reported by manufacturer (+/- 0.0007 g)

Calculations:

RSD(%) = $\frac{SD \times 100\%}{\bar{x}}$

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Data Sheet 1. Station 1.

Lavoisier Analytical Balance

SD

_______(___)

→

The mass for exactly one ounce Lavoisier would have obtained based on his reported SD (units)

_____________(___)

→

%RSD

____

↓

Mass +/- %RSD

______+/-_____(___)

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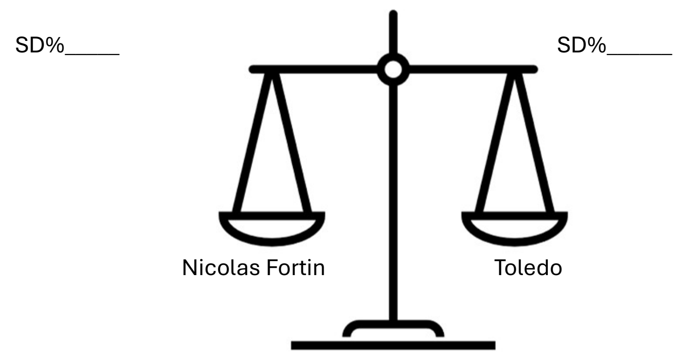
Data Sheet 1. Station 1.

Which balance has less uncertainty, Nicolas Fortin's or Toledo balance? Use SD% to compare the two balances.



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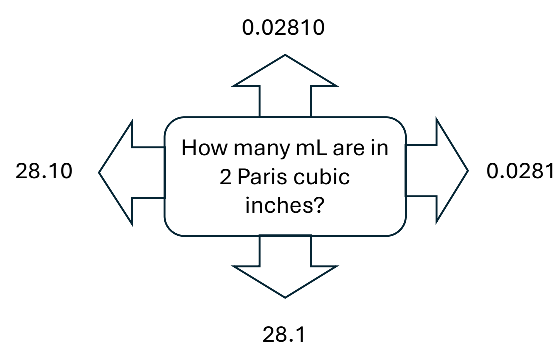
Data Sheet 2. Station 2

Calculate how many mL are in 2 Paris cubic inches and report it with the correct number of sig figs.

1 cubic foot = 24.28 L
2 Paris cubic inches or 1/864 cubic foot

Calculations:



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Data Sheet 2. Station 2

Graduate Cylinder, 50 mL

How many sig figs can you report based on the cylinder's smallest division?

____

Partner 1: Volume of measured 2 Paris cubic inches of water in mL

____

Partner 1: reading error (units)
+/-________(____)

Partner 2: Volume of measured 2 Paris cubic inches of water in mL

____

Partner 2: reading error (units)
+/-________(____)

Bystander: Volume of measured 2 Paris cubic inches of water in mL

____

Bystander: reading error (units)
+/-________(____)

Average (mean) volume, mL

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Data Sheet2. Station 2

Graduate Cylinder, 50 mL

Partner 1: Relative reading error (%)
+/-________%

Partner 2: Relative reading error (%)
+/-________%

Bystander: Relative reading error (%)
+/-________%

Calculations:

Relative Reading Error(%) = $\frac{\delta x \times 100\%}{\bar{x}}$

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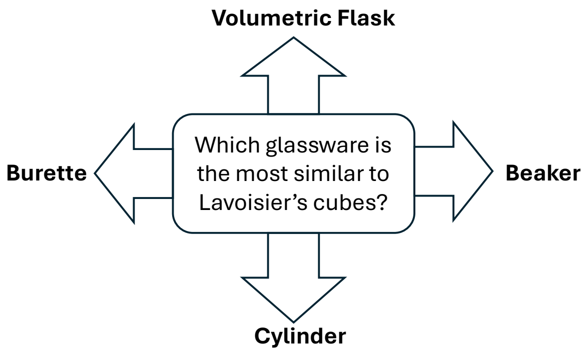
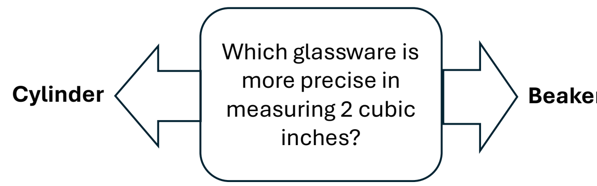
Data Sheet 2. Station 2

Look at the glassware available to you at this station and answer the following questions:





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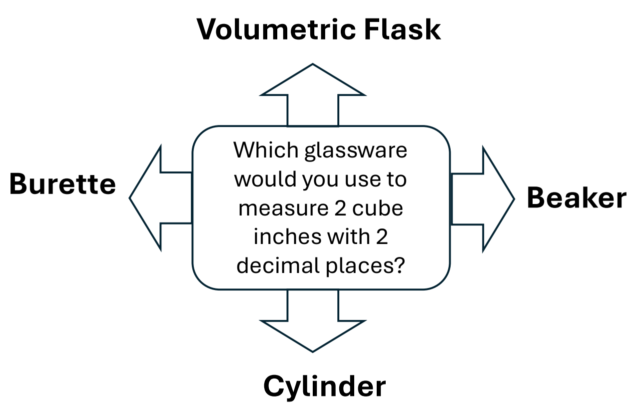
Data Sheet 2. Station 2

Look at the glassware available to you and answer the following question:



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Data Sheet 3. Station 3. Molarity Calculations

Calculate the number moles of NaCl in 1 ounce (average) you weighted. **<u>Hint</u>:** We will always take MW with 2 decimal places

Answer:____________________

Convert the volume of 2 cubic inches that you measured to L

Answer:____________________

Calculate molar concentration of NaCl in mol/L

Answer:________________

Definition of mole for Lavoisier's ghost:

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Data Sheet 4. Station 4. Spanish doubloons – gold or not?

1st trial: Mass of displaced water (units)
__________(_____)

2nd trial: Mass of displaced water (units)
__________(_____)

3rd trial: Mass of displaced water (units)
__________(_____)

Average (units)
___________(______)

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Data Sheet 4. Station 4. Spanish doubloons – gold or not?

Water temperature (units):
__________(_______)

Water density at this temperature (units):
_____________(_____)

Average mass of displaced water (units):
__________(_______)

Mass of dry coin (units):
__________(_____)

Coin density (units):
_______(______)

Material
____

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Data Sheet 5. Station 5.

Type of thermometer	Temperature, °C	Temperature, K
Analogue		
Digital		

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Data Sheet 6. Station 6.

- Please identify the statement that is written on the puzzle
- ____
- Give an example of experiments Lavoisier used to arrive to this statement?
- ____
  ____

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